

NATHAN DRESSLER - CURRICULUM VITAE

RESEARCH INTERESTS

String Theory & Quantum Gravity, the Swampland Program, Bordism/Cobordism, Generalized Symmetries, SCFTs

EDUCATION

University of Pennsylvania

- Ph.D. Physics and Astronomy (Aug. 2026 - Present)
 - Advisor: Prof. Jonathan Heckman

Virginia Tech

- Physics, B.S. (May 2026)
- Mathematics, B.S. (May 2026)
- GPA: 4.0 on 4.0 scale

EXPERIENCE & RESEARCH

- String Theory & Quantum Gravity Research under Dr. Jonathan Heckman (Aug. 2026 - Present)
 - Studying how bordism groups constrain non-perturbative structure of string/M-theory in the spirit of the Swampland Cobordism Conjecture
 - Computing $\text{Pin}^\pm$ and Spin bordism groups twisted by duality bundles to identify the reflection branes and other singular objects required to cancel candidate discrete global symmetries of M-theory and type IIB string theory
 - Investigating generalized (higher-form and non-invertible) symmetries of SCFTs engineered via M-theory on non-compact internal six-manifolds, using wrapped branes and Symmetry TFTs to extract defect groups and global structure
 - Exploring whether stable non-supersymmetric 4d CFTs can arise from M-theory compactified on non-Calabi-Yau internal geometries
- String Theory Research under Dr. Lara Anderson & Dr. James Gray (May 2025 – May 2026)
 - Explored the structure of 4-dimensional, $N=1$ heterotic string compactifications via complete intersection Calabi-Yau (CICY) threefolds
 - Studied flop transitions between CICY threefolds using GLSM/GIT wall-crossing methods
 - Explicitly constructing the flop of the CICY $P1[1,1] \times P4[1,4]$ to a twisted projective bundle over P^4
 - Reformulated flopped geometries in a non-negatively-twisted “total space” presentation, via $\text{GL}(2, \mathbb{Z})$ row operations on the GLSM charge matrix, to enable direct intersection-number and cohomology computations
 - Modeled the conifold-type nodes connecting pairs of CICY resolutions to a shared singular limit (e.g. the nodal quintic threefold $P4[5]$) using local linear sigma models, and identified the FI-parameter wall-crossing that realizes the flop
 - Diagnosed and resolved a subtlety in identifying the correct Kähler-cone wall corresponding to a genuine flop transition, as opposed to a divisorial contraction, via the secondary-fan/GIT chamber structure of the ambient toric varieties
- Algebraic Geometry Research under Dr. Leo Herr (November 2025 – May 2026)
 - Studying toric varieties as a combinatorial framework for algebraic geometry
 - Investigating resolution of singularities via toric methods, with emphasis on surface singularities
 - Constructing and analyzing fans corresponding to blowups and subdivisions of singular varieties
 - Computing intersection numbers and tracking divisor behavior under toric blowups
 - Developing explicit examples of singularity resolution using fan refinements
 - (optional stronger version if accurate later)
 - Applying toric techniques to explicitly resolve singularities via star subdivisions of cones
- Dark Matter Research under Dr. Bruce Vogelaar (Jan. 2024 - Present)

- Developed automated optical wire tension analyzer to ensure planarity of 1236 tensioned wires situated in a liquid argon environment under an electric field
- Analyzed quantitative changes in stainless-steel extension springs under cryogenic conditions to understand length contraction and young's modulus effects on spring constants
- Contributed to construction of automated spring-maker for DarkSIDE-20k's wire grid
- Contributed to construction of wire grid for DarkSIDE-20k dark matter detector
- Physics Data Analysis Research under Dr. Camillo Mariani (Jan. - May 2024)
 - Assisted implementation of neutron detector calibrating system
 - Used ROOT for boron capsule nuclear recoil data analysis
 - Performed silicon photomultiplier calibration and implementation for data collection
- Radiation Physics Intern at Virginia Cancer Specialists (Dec. - Jan. 2022/2023/2024, May - Aug. 2023)
 - Developed four Python scripts to increase efficiency of LINAC quality assurance tasks
 - Automated image & data analysis methods; decreased labor from five days to two hours
 - Studied nuclear physics extensively under Dr. Youssef Charara

TECHNICAL REPORTS & PAPERS

- Dressler, N., & Lai, E. (2025). *Automated frequency-based tension measurement of vibrating wires using video analysis and FFT* (Unpublished)
- Dressler, N., & Rhudy, I. (2024). *Properties of 304 stainless steel extension springs in cryogenic environments* (Unpublished)

POSTERS & TALKS

- Dressler, N., & Lai, E. (2025). *Construction of an automated optical wire tension analyzer*. Poster internally presented to DarkSIDE-20k collaboration
- Dressler, N., & Rhudy, I. (2024, July). *Grid-wire tensioning for dark matter TPC at cryogenic temperatures*. Poster presented at Virginia Tech REU Research Poster Session, Blacksburg, VA
- Dressler, N. (2024, April). *Calibration of low energy neutron detector*. Poster presented at Virginia Tech Undergraduate Research Poster Session, Blacksburg, VA

SCHOLARSHIPS & AWARDS

- NSF Graduate Research Fellowship (2026-2031)
- H.Y. Loh Award (Physics Valedictorian) (Spring 2026)
- Robert C. Richardson Scholarship (2023-2024)
- Richardson, Webster, and Sarah Scholarship (2024-2025, 2025-2026)
- Hamilton Prize (2025-2026)
- 1st Place Undergraduate Research Poster Award (Spring 2024, Summer 2024)
- Dean's List (Fall 2022/2023/2024, Spring 2023/2024/2025)

PROFESSIONAL SOCIETIES

- Phi Beta Kappa (National Honor Society, Apr. 2025 - Present)
- Sigma Pi Sigma (Physics Honors Society, Apr. 2025 - Present)